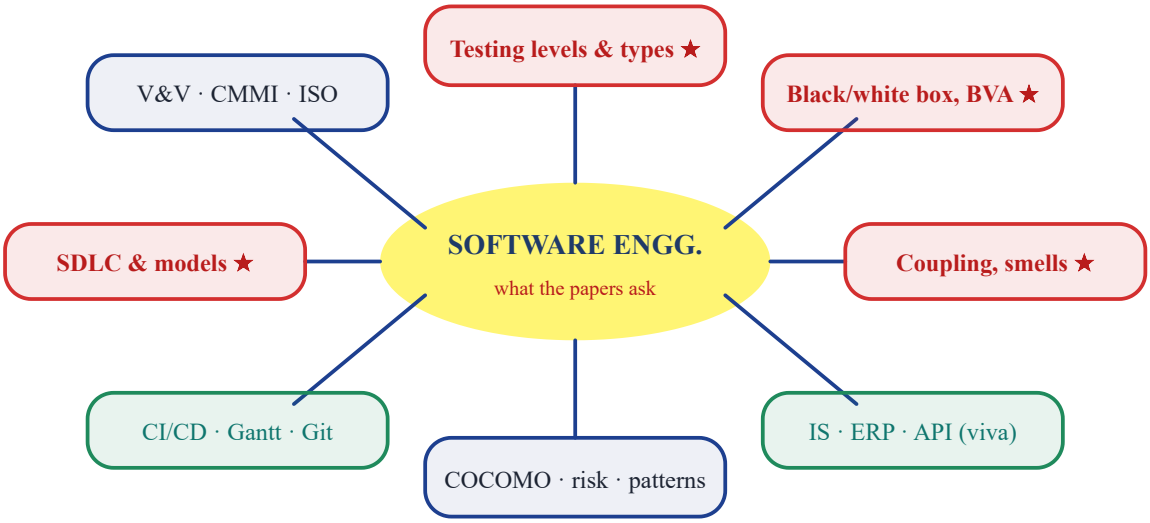
# Solved Past Questions: Software Engineering

**45 questions from BCS, bank, Petrobangla and BUET-set papers, solved in 12 sheets**




↑ the asked topics on one page: red = asked most often, blue = written once, green = MCQ / viva

## Papers covered

- 38th BCS Written (2018) · 49th BCS ICT paper 2, MCQ (2025)
- Bangladesh Bank AME (BIBM, 2023) · bank Assistant Programmer
- Petrobangla AM (IT) 2022 (BUET), AM (CSE) 2024 · BOGMC AM 2024
- Islamic Bank PLC SQA 2025 · Bangladesh Satellite Co. AE 2025, 2026 · bank IT viva boards

**Most asked**

① name the testing type
② black-box vs white-box
③ EC + BVA test cases
④ SDLC phases (10 marks)
⑤ coupling & cohesion
⑥ COCOMO, 32 KLOC

**Legend:** **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked answer · **blue frame** = the question as asked

## Contents


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## 1 Levels of Software Testing

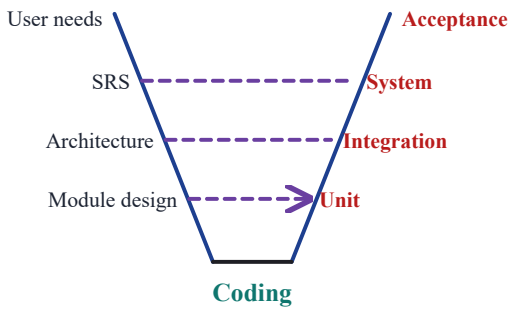
**Viva:** What are the levels of software testing? What is unit / integration / system / acceptance testing?

bank IT viva boards (asked one by one and again as "levels")

| Level | What is tested | By · against |
|---|---|---|
| **Unit** | one function / class alone (stubs, drivers, mocks) | developer · module design |
| **Integration** | interfaces and data flow between modules | developer, tester · architecture |
| **System** | whole system, functional **and** non-functional | independent testers · SRS |
| **Acceptance** | fit for business use: UAT, alpha, beta | client, users · business needs |

**Integration strategies:** **big-bang** (all at once) · **top-down** (needs **stubs** for lower modules) · **bottom-up** (needs **drivers** for upper modules) · **sandwich** (both). **Alpha** = users test at the developer's site; **beta** = real users at their own site.

**Memory trick: V-model:** each test level checks what the document opposite it promised: module design ↔ unit, architecture ↔ integration, SRS ↔ system, user needs ↔ acceptance.




V-model: left arm builds, right arm tests

## 2 Name the Testing Type: Functional or Not?

**Q4(b):** Identify the testing type in each scenario and say whether it is Functional, Non-Functional or Both: (1) verifying the smallest testable unit; (2) re-running tests to check the application still works after code changes; (3) performance and responsiveness under a specific amount of traffic; (4) preliminary testing of basic functions before detailed testing; (5) verifying that integrated modules work together.

Petrobangla AM (CSE) · 31.05.2024 · BOGMC (Petrobangla) AM · 2024 · also set in another recruitment paper

| # | Testing type | Why | Category |
|---|---|---|---|
| 1 | **Unit** | smallest testable part, in isolation | Functional |
| 2 | **Regression** | re-runs the existing suite after a change | **Both** |
| 3 | **Load** (performance) | behaviour under the expected traffic | Non-functional |
| 4 | **Smoke** (build check) | shallow check of a new build's key functions | Functional |
| 5 | **Integration** | interfaces between modules | Functional |

**Functional vs non-functional:** **Functional** = **what** the system does (input → expected output). **Non-functional** = **how well**: performance, load, stress, security, usability, reliability.

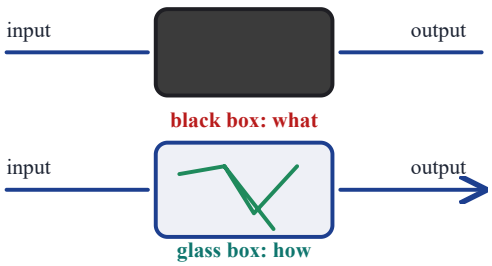
**Exam trap:** Regression re-runs functional **and** non-functional tests (a release can also get slower), so "Both". Re-testing re-runs only the failed case. Smoke = broad, shallow; sanity = narrow, deep.

## 3 Black-box vs White-box Testing

**Q6:** Differentiate between Black Box Testing and White Box Testing.

Islamic Bank PLC, SQA Senior Officer (Network) · 14.03.2025 · viva: "What are software testing methods?"

| Point | Black-box | White-box |
|---|---|---|
| Code knowledge | none: inputs and outputs only | full: internal logic and paths |
| Tests based on | requirements, SRS | source code, design |
| Also called | behavioural, closed-box | **structural**, glass-box, clear-box |
| Done by · level | testers, users · system, acceptance | developers · unit, integration |
| Techniques | EP, BVA, decision table, state transition | statement/branch/path coverage, basis path |
| Finds | missing or wrong functions, UI errors | logic errors, dead code, unused variables |




**Grey-box:** **partial** knowledge of the inside (DB schema, architecture, API docs) while testing from outside. Used in integration, web and **penetration testing**. Methods = black, white, grey box.

**Memory trick:** **Black**: you can't see in → test the **B**ehaviour. **White**: glass walls → test the **W**iring (code paths).

## 4 White-box = Structural Testing

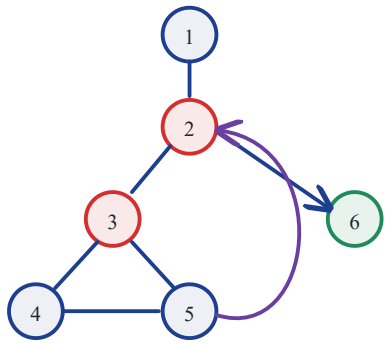
**Asked MCQs ✍️:** **Q86** Which technique is called structural testing? → **White box**. **Q21** Which is **not** a focus of white-box testing: code coverage / data flow / user perspective / code logic? → **User perspective** (that is black-box). 49th BCS ICT paper 2 · 2025

**Coverage, weakest → strongest:** statement → branch (decision) → condition → MC/DC → **path**. 100% branch ⇒ 100% statement, not the reverse.

> **Cyclomatic complexity:** **V(G) = E − N + 2** = decisions + 1 = regions = number of basis paths = minimum test cases for basis-path testing.

```
int f(int x){ int s = 0;
  while (x > 0) {       // decision 1
    if (x % 2) s += x;   // decision 2
    x--; }
  return s; }
```

Tests x = 0 (loop skipped), x = 1 (odd branch), x = 2 (even, then odd) cover every edge: **3 cases = V(G)**.




N = 6, E = 7

**V(G) = 7 − 6 + 2 = 3**

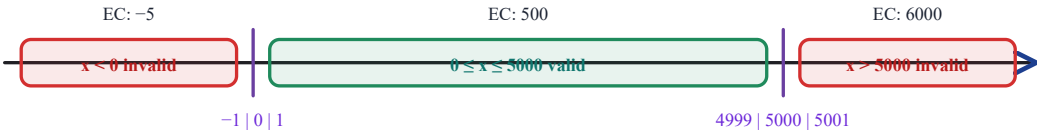
1 s=0 · 2 while · 3 if · 4 s+=x · 5 x-- · 6 return

## 5 Equivalence Classes & Boundary Values

> **Q3:** What is black box testing? A program computes the square root of an input integer between 0 and 5000. Determine the equivalence class test cases, and the test cases using boundary value analysis.

Petrobangla AM (IT) · 16.09.2022 (BUET)

**Black-box testing** tests the program only through its inputs and outputs, against the specification, without looking at the code (behavioural testing).




**Equivalence classes:** one representative per class → **{−5, 500, 6000}**. Every value in a class should behave alike.

**Boundary values:** errors cluster at the edges 0 and 5000 → **{−1, 0, 5000, 5001}**; the robust set adds 1, 4999 and a nominal 2500.

| Input | Expected output |
|---|---|
| −1, 5001 | rejected: "invalid input" |
| 0 · 1 · 2500 | 0 · 1 · 50 |
| 4999 · 5000 | 70.70 · 70.71 |

> **Exam trap:** A test case is an input **plus its expected output**: give both. For n inputs, normal BVA needs **4n + 1** cases (min, min+, nominal, max−, max), robust BVA **6n + 1** (adds min− and max+): here 5 and 7.

## 6 Verification & Validation (CMMI Level 3)

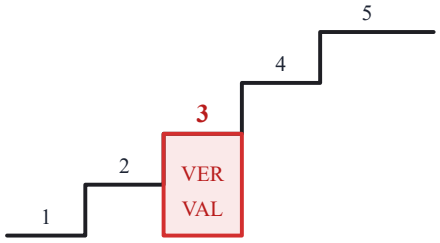
> **Q9:** Verification and validation are two process areas at CMMI level 3. For both: (a) give a definition; (b) describe how you can fulfil these areas in your software testing activities.

Bangladesh Bank, Assistant Maintenance Engineer · 04.02.2023 (BIBM)

**(a) Verification (VER):** ensures work products **meet their specified requirements**: "Are we building the product **right**?" Peer reviews, inspections, walkthroughs, static analysis, tests against the spec.

**(a) Validation (VAL):** shows the product **fulfils its intended use** in its real environment: "Are we building the **right** product?" UAT, alpha/beta, prototypes, usability tests.

1. **Plan:** pick work products and methods; build a traceability matrix (requirement → test cases)
2. **VER:** review SRS, design and test cases; code review, static analysis; unit, integration and system tests against the spec
3. **VAL:** users set acceptance criteria; UAT in a production-like environment; beta or pilot run
4. **Analyse:** log defects, find root causes, fix, re-test, regression; report coverage, defect density; sign-off




CMMI 1 Initial · 2 Managed · 3 Defined · 4 Quant. Managed · 5 Optimizing

## 7 The Software Development Life Cycle

**Q2:** Explain the concept of the Software Development Life Cycle (SDLC) and describe its main phases. 10 marks

bank Assistant Programmer, written

**Concept:** SDLC is a **structured framework of phases** to plan, build, test, deploy and maintain software that meets its requirements on time and within budget. Each phase has set activities and a deliverable that feeds the next; process models (waterfall, incremental, spiral, agile) differ in how they order and repeat the phases.

| Phase | Key activities | Output |
|---|---|---|
| 1 Planning, feasibility | scope, cost, schedule; technical, economic, operational feasibility | feasibility report |
| 2 Requirements | interviews, questionnaires, JAD sessions, prototypes | **SRS** |
| 3 Design | architecture (HLD), modules (LLD), DB, UI; DFD, ERD, UML | design document |
| 4 Coding | code to standards, version control, unit tests | source code |
| 5 Testing | integration, system and acceptance tests; fix defects | test report |
| 6 Deployment | install, convert data, train users, change over | live system |
| 7 Maintenance | corrective, adaptive, perfective, preventive changes | new versions |

**Memory trick:** "**P**lease **R**ead, **D**esign, **C**ode, **T**est, **D**eploy, **M**aintain": P-R-D-C-T-D-M.

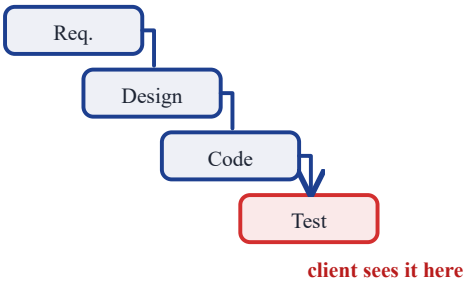
**Exam trap:** Over a product's life, maintenance is the **costliest** phase, and most of it is **perfective** (new features), not bug fixing.

## 8 System Implementation Stages

**Asked** in a recruitment paper: the stages of system development and implementation. 1 preliminary investigation (feasibility) → 2 analysis → 3 design → 4 coding → 5 testing → 6 **installation / implementation** → 7 maintenance. The implementation stage secures the users' **acceptance** of the system. **Changeover:** direct (cheapest, riskiest) · **parallel** (old and new together: safest, costliest) · pilot (one branch first) · phased (module by module).

## 9 Waterfall vs Incremental

| Point | Waterfall | Incremental |
|---|---|---|
| User feedback | only at testing | after each increment |
| Resources | tied to phases | one increment at a time |
| Requirements | frozen up front | can evolve |
| Risk | found late | cut early |

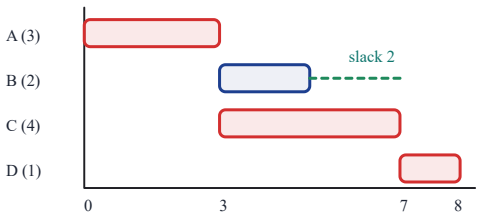



waterfall: one pass, no going back

> **Asked MCQ ✍️:** **Q40** Key disadvantage of the sequential waterfall model? → **Lack of client feedback**. Its phases do **not** overlap, and it is documentation-heavy, not lacking. 49th BCS · 2025

## 10 Project Scheduling & the Gantt Chart

> **Asked MCQs ✍️:** **Q9** Main goal of project scheduling? → **allocate resources and set deadlines**. **Q11** Which is used for project scheduling: pie chart / bar chart / Gantt chart / all? → **Gantt chart**. 49th BCS · 2025




Gantt chart in weeks; red = critical path A–C–D

**Gantt chart:** tasks as horizontal bars on a time axis, showing start, duration, overlap and progress. It shows **dependencies** poorly, so pair it with a **PERT/CPM network**.

**Critical path** = longest path = shortest project time; its tasks have **zero slack**. B and C need A, D needs B and C: A–C–D = 3 + 4 + 1 = **8 weeks**; B may slip 2 weeks.

> **Exam trap:** A Gantt chart is drawn with bars, but the standard answer is **Gantt chart**; a pie chart shows shares, not time, so "all" is wrong.

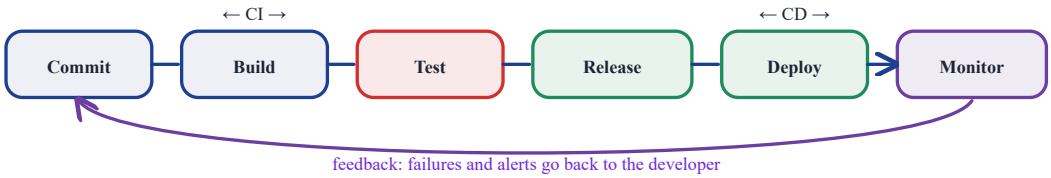
## 11 CI/CD: Stages & Benefits

> **Q7:** What is the CI/CD development model? Explain its major stages and benefits.

Islamic Bank PLC, SQA Senior Officer (Network) · 14.03.2025

**CI (continuous integration):** developers merge small changes into the shared main branch **several times a day**; every merge triggers an automatic build and test run. **CD** = **continuous delivery** (every green build is releasable; going live needs one

manual approval) or **continuous deployment** (it goes live automatically).




1. **Source:** a push or pull request starts the pipeline
2. **Build:** compile, fetch dependencies, make an artifact or container image
3. **Test:** unit and integration tests, static analysis, security scans
4. **Release:** deploy to staging; smoke, acceptance, performance tests
5. **Deploy:** to production by rolling, blue-green or canary release
6. **Monitor:** logs, metrics, alerts, rollback

**Benefits:** bugs caught minutes after they are written · small, low-risk releases · faster delivery · repeatable builds, fewer manual errors · quick rollback · constant feedback. Tools: Git, Jenkins, GitHub Actions, GitLab CI, Docker, Kubernetes.

**Git: commit vs push:** **commit** saves a snapshot in your **local** repository (works offline); **push** uploads local commits to the **remote** so the team and CI see them. bank Assistant Programmer, Q5

**Memory trick:** **add** → stage · **commit** → my desk · **push** → the team's shelf · **pull** = fetch + merge. CI = "merge often, test always"; CD = "always ready to ship".

## 12 Coupling & Cohesion

**Q3:** Explain the concepts of Coupling and Cohesion with examples.

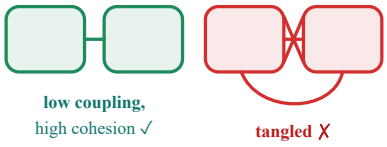
Bangladesh Satellite Company PLC, Assistant Engineer (CSE) · 23.08.2025

**Coupling** = how much modules **depend on each other**; aim **low**, so a change does not ripple. **Cohesion** = how strongly the parts **inside one module** serve one task; aim **high**, so the module is easy to understand, test and reuse.

| Coupling, best → worst | Example | Cohesion, best → worst | Example |
|---|---|---|---|
| Data | area(w, h) | Functional | computeTax(): one job |
| Stamp | whole record, one field used | Sequential | read → validate → save |
| Control | print(rep, flag) | Communicational | print + update one record |
| External | share a file format | Procedural | steps in order, other data |
| Common | same global variable | Temporal | init() at start-up |
| Content | edits another's internals | Logical | doIO(type) by flag |
| | | Coincidental | Utils of unrelated code |

**Memory trick:** Coupling: "**D**o **S**weet **C**hildren **E**at **C**old **C**akes?" Cohesion: "**F**un **S**equels **C**ommunicate **P**lots **T**hat **L**ook **C**oincidental".

**Exam trap:** Good design = **low coupling, high cohesion**. "High coupling" is never a design principle.



## 13 Code Smells: Examples & Fixes

**Q11:** Give examples of the following code smells: (a) Feature Envy (b) Dead Code (c) Duplicate Code (d) Shotgun Surgery.

asked in a recruitment paper

**(a) Feature Envy**: a method uses another object's data more than its own.

```
class ShippingCalc {
 double cost(Order o) {
  return o.getWeight()
       * o.getZone().getRate(); } }
```

Fix: **Move Method** → Order.shippingCost().

**(b) Dead Code**: never runs, or its result is never used.

```
double price(double p) {
  double old = p * 0.95; // never used
  if (LEGACY_V1) { ... } // always false
  return p * 0.90; }
```

Fix: delete it; version control keeps the history.

**(c) Duplicate Code**: the same logic pasted in several places.

```
// in register() AND updateProfile()
if (email == null
    || !email.contains("@"))
  throw new IllegalArgumentException();
```

Fix: **Extract Method** validateEmail().

**(d) Shotgun Surgery**: one change forces small edits in many classes.

```
// VAT 0.15 typed into Cart, Invoice,
// Receipt and Report: new rate = 4 edits
```

Fix: one TaxPolicy class or constant (Move Field).

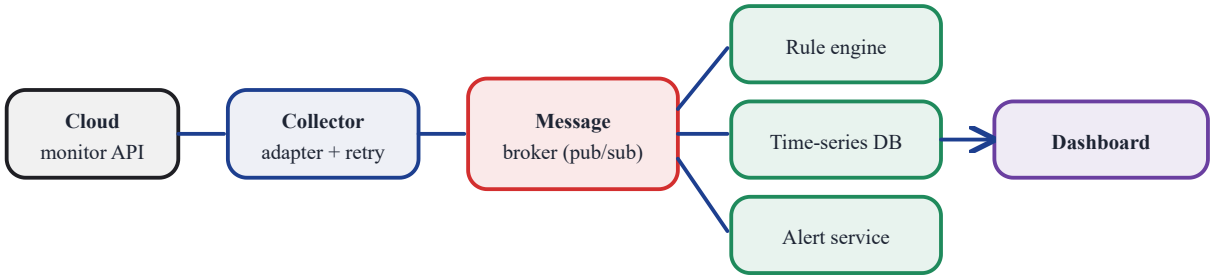
**Exam trap:** **Shotgun surgery** = one change → many classes. **Divergent change** is the opposite: one class changed for many different reasons.

## 14 Architecture & Patterns: a Monitoring System

**Q2:** A third-party cloud service is used for continuous system monitoring through APIs. (a) Which software architecture would you recommend? (b) Which design patterns would be appropriate? (c) Explain why you selected them.

Bangladesh Satellite Company PLC, Assistant Engineer (CSE) · 21.08.2026

**(a) Architecture:** an **event-driven architecture** of small services (microservices; or modules of one app for a small team), with a **message broker** in the middle and an **adapter layer** facing the third-party API.




poll the API or receive webhooks → publish events → independent subscribers

| (b) Pattern | Where it is used |
|---|---|
| **Adapter** / Facade | wrap each vendor API behind our own MetricsSource interface |
| **Observer** / Publish–Subscribe | rule engine, alerting and dashboard subscribe to metric events |
| **Circuit Breaker** + Retry | vendor outages, timeouts, rate limits (exponential backoff) |
| **Strategy** · Factory | swappable alert rules and channels · a collector per provider |

**(c) Justification:** ① **Loose coupling to the vendor:** the adapter isolates API changes; a new provider touches one class. ② **Scalability:** collectors and processors scale separately; the broker absorbs bursts. ③ **Reliability:** the circuit breaker and the queue stop one failure from cascading, and no event is lost. ④ **Real-time:** events are pushed to subscribers at once. ⑤ **Extensibility:** a new alert channel is one more subscriber or strategy (open–closed principle). ⑥ **Testability:** every part can be tested against a mock API.

**Exam trap:** State the trade-off: microservices add deployment and monitoring overhead; for a small team a modular monolith with the same patterns is a valid answer. The justification earns more marks than the label.

## 15 User Interface Design

**Asked MCQ ✍️:** **Q72** Which principle is followed in UI design: high coupling / user-centred design / sequential processing / cost optimization? → **User-centred design**. 49th BCS · 2025

**User-centred design:** design around real users' goals and tasks: involve users early, test usability with them, **iterate**. Golden rules: consistency, feedback, error prevention, easy undo, user in control, low memory load.

## 16 COCOMO: Development Time

**Q96:** An organic software has 32 KLOC. The nominal development time (in months) using the COCOMO model is: (a) 6 (b) 7 (c) 14 (d) 21

49th BCS ICT paper 2 · 2025

**Basic COCOMO (Boehm, 1981):**

Effort $\mathbf{E = a(KLOC)^b}$ person-months

Time $\mathbf{T = c(E)^d}$ months

Staff = E / T · productivity = LOC / E

| Mode | a | b | c | d |
|---|---|---|---|---|
| Organic | 2.4 | 1.05 | 2.5 | 0.38 |
| Semi-detached | 3.0 | 1.12 | 2.5 | 0.35 |
| Embedded | 3.6 | 1.20 | 2.5 | 0.32 |

**Solution ✍️:**

1. $32^{1.05} = 38.05$
2. $E = 2.4 \times 38.05 = \mathbf{91.3\ PM}$
3. $91.3^{0.38} = 5.56$
4. $T = 2.5 \times 5.56 = 13.9 \approx \mathbf{14\ months} \rightarrow \mathbf{(c)}$
5. Staff = 91.3 / 13.9 ≈ 6.6 ≈ 7 people; productivity ≈ 350 LOC/PM
6. Same 32 KLOC: semi-detached 145.5 PM, embedded 230.4 PM, both ≈ 14.3 months

**Exam trap:** Intermediate COCOMO uses a = **3.2** for organic (then × EAF from 15 cost drivers). Putting 3.2 into the basic formula gives E ≈ 121.8 PM and T ≈ **15.5** months, not 14. Option (b) 7 is the **staff** figure, not months.

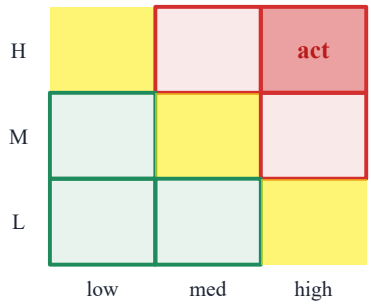
## 17 Risk Analysis Techniques

**Q9(a):** Describe the different techniques used in risk analysis of software development.

38th BCS Written · 2018

Risk management = **identify → analyse → prioritise → plan → monitor**. The analysis techniques:

1. **Identification:** risk checklists, brainstorming, Delphi (anonymous experts), interviews, SWOT, assumption analysis, past-project lessons
2. **Categorising:** project, product, business risks; or five impact areas: new technology, requirements, architecture, performance, organisation
3. **Qualitative:** rate probability and impact (negligible → catastrophic) on a **probability–impact matrix**
4. **Quantitative:** risk exposure **RE = P × L**; decision trees (expected value), sensitivity analysis, Monte Carlo simulation
5. **Prioritise & respond:** rank by RE; avoid, mitigate, transfer or accept (RMMM plan)




probability ↑ × impact →

| Worked example | P | Loss L | RE = P × L | Rank |
|---|---|---|---|---|
| requirements keep changing | 0.6 | Tk 5 lakh | Tk 3.0 lakh | 1 |
| key developer leaves | 0.3 | Tk 6 lakh | Tk 1.8 lakh | 2 |
| third-party API changes | 0.5 | Tk 2 lakh | Tk 1.0 lakh | 3 |

**Exam trap:** The question asks for **techniques**. Listing only risk categories (technology, requirements …) earns part marks; add identification methods, the matrix and RE = P × L.

## 18 Viva Rapid-Fire

**Q. What is an API?**
→ Application Programming Interface: a **contract** (endpoints, formats, rules) that lets one program use another's services or data without knowing its internals.

**Q. How does an API work?**
→ The client sends a request (HTTP method, URL, headers with an auth token, JSON body) → the API authenticates, runs the logic, reads the DB → returns a status code (200, 201, 400, 401, 404, 500) and JSON. Styles: REST, SOAP, GraphQL, gRPC.

**Q. What is an Information System?**
→ **people, hardware, software, data and procedures** (plus

**Q. What is ERP? Why is it used?**
→ Enterprise Resource Planning: one integrated suite on a **single database** for finance, HR, procurement, inventory, sales, production. Gives one source of truth, no double entry, automated workflows, real-time reports, an audit trail.

**Q. What is MRP? MRP vs ERP?**
→ Material Requirements Planning: from the production schedule, bill of materials and stock it decides **what material, how much, when**. MRP = manufacturing materials only; MRP II added capacity and finance; **ERP = the whole enterprise**.

network) that collect, process, store and distribute information.

**Q. Role of IS in an organisation?**
→ runs daily transactions (TPS), supports decisions (MIS, DSS, ESS), communication, records and compliance, competitive advantage.

**Q. Management software? Examples?**
→ software to plan, run and control resources and processes: ERP (SAP, Oracle, Microsoft Dynamics 365, Odoo), Tally (accounting), CRM, HRM/payroll, Jira; in banks, core banking (Temenos T24, Finacle).

**Q. Do you know app development?**
→ Android: Kotlin/Java; iOS: Swift; cross-platform: Flutter (Dart), React Native; the app talks to the server through REST APIs. Name an app you built.

**Q. System vs solution?**
→ **System**: interacting components doing one function (a core banking system). **Solution**: the full answer to a business problem: one or more systems plus process change, integration, training, support.

**Q. How is ISO applied in software?**
→ ISO 9001 (quality management), ISO/IEC 12207 (life-cycle processes), ISO/IEC 25010 (product quality), ISO/IEC/IEEE 29119 (testing), ISO/IEC 27001 (security); through defined processes and certification audits.

**Q. Levels and methods of testing? Acceptance testing?**
→ unit, integration, system, acceptance (sheet 2); black, white, grey box (sheet 3). Acceptance = users confirm business fit before go-live.

## 19 All MCQs Asked, at a Glance

| Q | Asked | Answer | Why |
|---|---|---|---|
| 9 | goal of scheduling | resources and deadlines | who does what, when |
| 11 | scheduling tool | Gantt chart | tasks on a time axis |
| 21 | NOT a white-box focus | user perspective | that is black-box |
| 40 | waterfall's drawback | no client feedback | client sees it at the end |
| 72 | UI design principle | user-centred design | design around users |
| 86 | "structural testing" | white box | tests code structure |
| 96 | organic 32 KLOC time | **14 months** | $2.5 \times 91.3^{0.38}$ |

# Night-Before-the-Exam Sheet

**Formulas & numbers:**

$E = a \cdot KLOC^b$ · $T = 2.5 \cdot E^d$ · staff = E/T

organic 2.4/1.05/0.38 · semi 3.0/1.12/0.35

embedded 3.6/1.20/0.32 · intermediate a = 3.2

32 KLOC organic → 91 PM, 14 months, ≈ 7 people

V(G) = E − N + 2 = decisions + 1

BVA: 4n + 1 cases · robust: 6n + 1

RE = probability × loss

critical path = longest path, zero slack

CMMI: 1 Initial · 2 Managed · 3 Defined

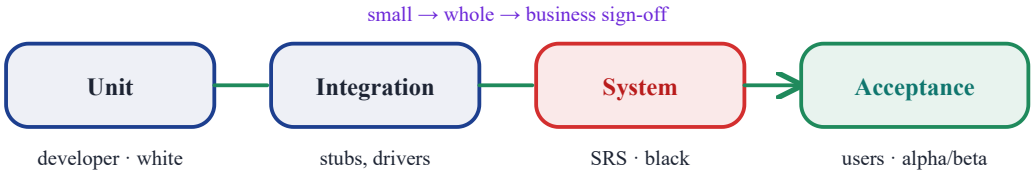
4 Quantitatively Managed · 5 Optimizing

## One-liners that were asked

1. Levels: unit → integration → system → acceptance
2. Unit = developer; UAT = users (alpha, beta)
3. Regression = both; load = non-functional
4. Smoke = shallow check of a new build
5. White box = structural = glass box
6. Black box = behavioural, the user's view
7. Grey box = partial knowledge (pen-testing)
8. Verification = product right; validation = right product
9. VER and VAL: CMMI level-3 process areas
10. Waterfall flaw: no client feedback till the end
11. Changeover: direct, parallel, pilot, phased
12. CI = merge often, auto build/test; CD = always releasable
13. commit = local, push = remote
14. Low coupling (data best), high cohesion (functional best)
15. UI design = user-centred design
16. ERP = one database for every department

**Top traps:** regression is "both" · a test case needs its expected output · BVA values sit on and just outside each edge · intermediate a = 3.2 gives 15.5 months, not 14 · 7 in the COCOMO MCQ is staff, not months · waterfall phases do not overlap · shotgun surgery ≠ divergent change · risk "techniques" ≠ risk categories

**All tricks together:** V-model: each test level checks the document opposite it · Black = Behaviour, White = Wiring · P-R-D-C-T-D-M for SDLC · add → stage, commit → my desk, push → team shelf · "Do Sweet Children Eat Cold Cakes?" (coupling) · "Fun Sequels Communicate Plots That Look Coincidental" (cohesion)




the four levels in one line

## ✓ Practice MCQs (exam pattern, 10 marks)

1. Basic COCOMO effort for 32 KLOC, organic:
   (a) 46 PM (b) 91 PM
   (c) 122 PM (d) 146 PM
2. Which testing type is non-functional?
   (a) Smoke (b) Unit
   (c) Integration (d) Load
3. "Are we building the right product?" is:
   (a) Validation (b) Verification
   (c) Regression (d) Inspection
4. The most desirable coupling is:
   (a) Content (b) Common
   (c) Data (d) Control
5. Re-running old tests after a bug fix is:
   (a) Smoke (b) Regression
   (c) Alpha (d) Stress
6. A flow graph has 9 edges, 7 nodes. V(G) =
   (a) 4 (b) 2
   (c) 3 (d) 16
7. Input valid from 1 to 100: equivalence classes =
   (a) 1 (b) 2
   (c) 4 (d) 3
8. One change forces edits in many classes:
   (a) Feature envy (b) Dead code
   (c) Shotgun surgery (d) Duplicate code

**9.** Pattern that wraps a third-party API behind your own interface:

(a) Adapter (b) Singleton
(c) Observer (d) Strategy

**10.** VER and VAL are process areas at CMMI level:

(a) 2 (b) 5
(c) 4 (d) 3

Answer key → 1-b · 2-d · 3-a · 4-c · 5-b · 6-a · 7-d · 8-c · 9-a · 10-d

**Written question:** Using basic COCOMO, estimate the effort, development time, average staff and productivity of a 50 KLOC **semi-detached** project. How would the effort change if it were organic?

8 marks

1. Semi-detached: a = 3.0, b = 1.12, c = 2.5, d = 0.35; $E = a \cdot KLOC^b$, $T = c \cdot E^d$ 1
2. $50^{1.12} = 79.96 \rightarrow E = 3.0 \times 79.96 =$ **239.9 PM** 2
3. $239.9^{0.35} = 6.81 \rightarrow T = 2.5 \times 6.81 =$ **17.0 months** 2
4. Staff = 239.9 / 17.0 ≈ **14 people**; productivity = 50 000 / 239.9 ≈ **208 LOC/PM** 2
5. Organic: $E = 2.4 \times 50^{1.05} = 145.9$ PM, T = 16.6 months: about **40% less effort** for nearly the same time 1

## Before the exam ✓

- ✓ Can name any testing scenario: functional / non-functional / both
- ✓ Can write EC + BVA cases with expected outputs
- ✓ Can draw a flow graph and compute V(G)
- ✓ Can write the SDLC phases with outputs (10 marks)
- ✓ Can do COCOMO in all three modes

**Also practise (written):**

① V&V for CMMI level 3
② coupling & cohesion with code
③ an example of each code smell
④ CI/CD stages and benefits
⑤ risk analysis techniques